Relativistic Hills Mechanism

line element: $ds^2 = -\left(1-\frac{2M}{r}\right)dt^2 + \left(1+\frac{2M}{r}\right)^{-1}dr^2 + r^2 d\varphi^2$

(motion in $\theta = \pi/2$ plane)

conserved quantities:

$$\varepsilon = \left(1-\frac{2M}{r}\right)\frac{dt}{d\tau} \quad ; \quad \ell = r^2\frac{d\varphi}{d\tau}$$
$$= \gamma_\infty \qquad\qquad\qquad\qquad = b\,v_\infty$$

Anywhere along the trajectory, the energy measured by a static observer is

$$E = -\bar{p}\cdot\bar{u}_0 \quad ; \quad \bar{p} = m\bar{u} \text{ (particle)}$$

$\bar{u}_0 =$ 4-velocity of static observer

$$\bar{u}_0\cdot\bar{u}_0 = g_{tt}(u_0^t)^2 = -1 \qquad (u_0^i = 0)$$

$$\Rightarrow \quad u_0^t = \left(1-\frac{2M}{r}\right)^{-1/2}$$

$$\Rightarrow E = -m\,g_{tt}\,u^t u_0^t = m\underbrace{\left(1-\frac{2M}{r}\right)u^t}_{\varepsilon}\left(1-\frac{2M}{r}\right)^{-1/2}$$

$$\Rightarrow E(r) = m\left(1-\frac{2M}{r}\right)^{-1/2}\varepsilon$$

$$= m\,\gamma(r)$$

The distance $s$ of closest approach is given by

$$\varepsilon^2 = \cancelto{0}{\left(\frac{dr}{d\tau}\right)^2} + \left(1 - \frac{2M}{s}\right)\left(1 + \frac{\ell^2}{s^2}\right). \qquad \left(\begin{array}{c}\text{not necessary}\\ \text{to solve here}\end{array}\right)$$

$$E(s) = m\gamma_s .$$

At pericenter, give the particle a kick so that

$$\gamma_s \to \gamma_s + \delta\gamma$$

$$\Rightarrow E'(s) = m(\gamma_s + \delta\gamma)$$

$$= m\left(1 - \frac{2M}{s}\right)^{-1/2} \varepsilon'$$

$$\varepsilon' = \gamma'_\infty$$

$$\Rightarrow \gamma'_\infty = \left(1 - \frac{2M}{s}\right)^{1/2} (\gamma_s + \delta\gamma)$$

$$\left[\gamma = (1-v^2)^{-1/2} \Rightarrow \delta\gamma = (1-v^2)^{-3/2} v\,\delta v\right]$$

$$\gamma_s = \gamma_\infty \left(1 - \frac{2M}{s}\right)^{-1/2}$$

$$\Rightarrow \gamma'_\infty = \left(1 - \frac{2M}{s}\right)^{1/2}\left[\left(1 - \frac{2M}{s}\right)^{-1/2}\gamma_\infty + \delta\gamma\right]$$
$$= \gamma_\infty + \left(1 - \frac{2M}{s}\right)^{1/2}\delta\gamma$$

$$\Rightarrow \delta\gamma_\infty = \left(1 - \frac{2M}{s}\right)^{1/2}\delta\gamma$$

$$\delta\gamma = (1-v_S^2)^{-3/2} v_S \,\delta v = \gamma_S^3 \left(1-\frac{1}{\gamma_S^2}\right)^{1/2} \delta v$$

$$= \gamma_\infty^3 \left(1-\frac{2M}{S}\right)^{-3/2} \left[1 - \frac{1}{\gamma_\infty^2}\left(1-\frac{2M}{S}\right)\right]^{1/2} \delta v$$

$$[\,] = 1 - \left(\frac{1}{\gamma_\infty^2} - 1 + 1\right)\left(1-\frac{2M}{S}\right) = 1 - \left(1-\frac{2M}{S}\right) + \left(1-\frac{1}{\gamma_\infty^2}\right)$$

$$= \frac{2M}{S} + \underbrace{\left(1-\frac{1}{\gamma_\infty^2}\right)}_{v_\infty^2}$$

$$= \frac{2M}{S}\left[1 + \frac{v_\infty^2}{2M/S}\right]$$

$$\Rightarrow \delta\gamma_\infty = \left(1-\frac{2M}{S}\right)^{1/2} \gamma_\infty^3 \left(1-\frac{2M}{S}\right)^{-3/2} \left(\frac{2M}{S}\right)^{1/2} \left(1 + \frac{v_\infty^2}{2M/S}\right)^{1/2} \delta v$$

$$\delta\gamma_\infty = \gamma_\infty^3 \left(1-\frac{1}{\gamma_\infty^2}\right)^{1/2} \delta v_\infty = \ldots .$$

$$\Rightarrow \underbrace{\left(1-\frac{1}{\gamma_\infty^2}\right)^{1/2}}_{v_\infty} \delta v_\infty = \left(1-\frac{2M}{S}\right)^{-1} \left(\frac{2M}{S}\right)^{1/2} \left(1+\frac{v_\infty^2}{2M/S}\right)^{1/2} \delta v$$

$$\delta v_\infty^2 = 2\,\delta v \left(\frac{2M}{S}\right)^{1/2} \left(1-\frac{2M}{S}\right)^{-1} \left(1+\frac{v_\infty^2}{2M/S}\right)^{1/2}$$

$v_\infty^2 \ll 2M/S$

$$\Rightarrow v_\infty'^2 = \underbrace{2\,\delta v \left(\frac{2M}{S}\right)^{1/2}}_{\text{Newtonian result}} \overbrace{\left(1-\frac{2M}{S}\right)^{-1}}^{\text{GR correction}}$$

We made an <u>implicit</u> assumption that $\delta v_\infty$ is small by using the $O(\delta v_\infty)$ approximation for $\delta\gamma_\infty$.

This is okay for most cases, but we should be able to show $v_\infty' < 1$.

$$\Delta\gamma_\infty = \gamma_\infty' - \gamma_\infty = \gamma_\infty^3 \left(1 - \frac{2M}{s}\right)^{-1} \left(\frac{2M}{s}\right)^{1/2} \left(1 + \frac{v_\infty}{2M/s}\right)^{1/2} \delta v$$

$v_\infty \ll 1$ (neglect $O(v_\infty^2)$)

$$\Rightarrow \quad \gamma_\infty' - 1 = \delta v \left(\frac{2M}{s}\right)^{1/2} \left(1 - \frac{2M}{s}\right)^{-1}$$

$$\Rightarrow v_\infty'^2 = \left\{ 1 - \left[ \delta v \left(\frac{2M}{s}\right)^{1/2} \left(1 - \frac{2M}{s}\right)^{-1} + 1 \right]^{-2} \right\}$$

as $s \to 2M$, $\underline{\underline{v_\infty' \to 1}}$.